BAS-101- Physics

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MODULE -I

Frame of reference, Galilean transformation, Inertial and Non-inertial frames, Postulates of special theory of relativity, Michelson-Morley experiment, Lorentz transformation of space and time, Length contraction, Time dilation, Addition of velocities, Variation of mass with velocity, Equivalence of mass and energy, Momentum-energy transformation equations

Interference

Theory of interference of light, Conditions for sustained interference, Classification of interference, Fresnel‟s Biprism experiment, displacement of fringes, Interference in thin films-wedge shaped film and Newton‟s rings.

MODULE- II

Diffraction

Single, Double & N- slit Diffraction, Diffraction grating, Grating spectra, Rayleigh‟s criterion and resolving power ofgrating

Polarization

Phenomena of double refraction, Doubly refracting crystals, Quarter wave plate & Half wave plate, Nicol prism, Production and analysis of plane, circular and elliptical polarized light, Optically active substance, Fresnel‟s theory of optical activity, Specific rotation and Polarimeters.

MODULE -III

Laser and Holography

Spontaneous and stimulated emission of radiation, Einstein‟s coefficients, construction and working of Ruby, He-Ne lasers and laser applications,
Basic Principle of Holography, Construction and reconstruction of Image on hologram and applications of holography

Fiber Optics

Fundamental ideas about optical fiber, Types of fibers, Acceptance angle and cone, Numerical aperture, Propagation mechanism and communication in optical fiber, Attenuation, Signal loss in optical fiber anddispersion.

Reference Books:

1. *Aurthur Beiser, "Concepts of Modern Physics" - (Mc-GrawHill)*
2. *Robert Resnick – "Introduction to Special theory of Relativity" -Wiely*
3. *Ajoy Ghatak ," Optics - (TMH)" Brijlal & Subramanian (S. Chand)*
4. *Anuradha De., "Optical Fibre & Laser "- (New Age)*